# Cited but Not Correct: Auditing Evidence-Grounded LLM Configuration Advice against Equally Informed Baselines

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## Abstract

Language models increasingly recommend actions and justify them with citations to evidence in the prompt. Such a rationale makes three promises: the action is good, its claims about the evidence are true, and the decision rests on what is cited. Evaluations rarely test all three, or compare against what the evidence alone would recommend. Our audit protocol gives an equally informed non-LLM selector the same frozen evidence, ablates and deletes evidence, verifies every quantitative claim in an explanation against it, and scores decisions on independently labeled held-out outcomes. We apply it to configuration advice on real Linux kernel interventions with GroundKern, a testbed separating evidence, proposals and enforcement, and ten LLMs from 7B to frontier scale. The promises come apart. Evidence is necessary: without it, nine models mostly choose worse. It is also sufficient: with it, nine reproduce a table-lookup selector, which is optimal here. Citations are valid but one-sided: 252 of 260 cited runs belong to the chosen configuration. Explanations are cited but not correct: 41 of 97 audited claims misstate the evidence, mostly alongside a correct decision, and deletion reveals citation-specific reliance in one of five estimable models. Prompts, responses, annotations, code and logs accompany the paper.

## 1 Introduction

Language models are increasingly asked not only to answer questions but to recommend actions, and to justify each recommendation with citations to evidence supplied in the prompt. A rationale of this kind makes three distinct promises: that the recommended action is good, that what the rationale says about the evidence is true, and that the cited evidence is what the recommendation rests on. Evaluations usually check at most one of them. Work on attribution asks whether generated statements are supported by their sources (Rashkin et al., 2023; Gao et al., 2023b; Min et al., 2023; Liu et al., 2023), typically for question answering or summarization, where no downstream action is scored. Work on LLM decision-making scores the action (Yang et al., 2024; Lao et al., 2024; Giannakouris and Trummer, 2025) but treats the rationale as commentary. Neither asks what the evidence alone would have recommended.

**Choice:** `s50000_aall` (the best configuration) **Cited:** `training-table`, five retrieval runs (all valid IDs)

*"... For s50000_aall, the P95 latencies across 10 runs are consistently low: 126.36, 140.554, 94.81, ... 87.347 µs. The maximum among these is 183.217 µs, which is lower than the lowest P95 observed for any other config ..."*

**Check:** every quoted value is correct; the underlined comparison is false, since `s50000_aone` has a run at 85.033 µs.

Figure 1: A real full-context response from DeepSeek-V3.2 (seed 4089), abridged. The decision is correct and every citation is valid, yet the explanation misstates the evidence it cites. Across ten LLMs, 41 of 97 audited claims are misstated (§5).

We study these promises in configuration advice, an application where LLMs read manuals and measurements and propose database or kernel settings with natural-language justifications (Trummer, 2022; Lao et al., 2024; Giannakouris and Trummer, 2025; Chen et al., 2024; Lin et al., 2025). The setting is unusually well suited to auditing grounded rationales. The evidence is small, structured and numeric, so most claims about it can be checked exactly rather than by judging entailment. The quality of a decision can be measured on held-out runs, independently of the model and of any mechanism that enforces its advice. And the evidence can be handed unchanged to a conventional selector, which says what the evidence determines without a language model.

We ask three questions. *Added value*: does the model choose better than a conventional selector

that receives the same evidence, or does it only repackage that evidence? *Explanation correctness*: when an explanation cites valid evidence, does it state that evidence correctly? *Reliance*: does the cited evidence actually influence the decision? To answer them we build GroundKern, a testbed that separates evidence (a typed interaction graph and retrieved measurement traces), proposals (an LLM that emits a configuration with citations and an explanation) and enforcement (a runtime that validates, vetoes or stages the proposal). Because the evidence is an explicit, frozen object, it can be given to a non-LLM selector, ablated, or partially deleted; because enforcement is separate, its errors can be counted against labels computed from raw measurements.

Auditing ten LLMs, from 7B open-weight to hosted frontier models, on real Linux timer-slack and CPU-affinity interventions, we find that the three promises come apart. Evidence is necessary: without it, nine models mostly choose a worse configuration, and the local models we audited justify that choice by inventing meanings for opaque configuration names. It is also sufficient: with it, nine models reproduce the choice of a selector that simply reads the training table, a choice that is also best on the test grid. Citations are valid but one-sided and front-loaded: 252 of 260 cited retrieval runs belong to the chosen configuration, and 209 come from the first half of the evidence list. Explanations are cited but not correct (Figure 1): 16 of 28 audited claims from local models and 25 of 69 from hosted models misreport values, ranges or units, attribute a value to the wrong source, or overstate a regularity, and most of these accompany a correct decision. Deleting cited evidence reveals citation-specific reliance in only one of the five models for which a matched control exists.

**Contributions.** (i) An audit protocol for evidence-grounded LLM advice that combines a matched-information non-LLM baseline, evidence-source ablations, cited versus matched uncited deletion, claim-level verification of explanations against the supplied evidence, and outcome labels independent of enforcement, with all data roles frozen before test data are opened (§4). (ii) An audit of ten LLMs reported with every denominator, including an analysis of what models cite and a claim-level taxonomy of explanation errors (§5). (iii) The GroundKern testbed, whose calibrated risk gate provides a cautionary result on selective prediction under shift, with prompts, raw responses, per-claim annotations, code and measurement logs provided as supplementary material (§3).

The task is deliberately small, one workload family on one host, and its default configuration is best, so the audit can reveal failures of added value but not successes; we make no LLM-superiority, deployment-efficacy or production-safety claim (see Limitations).

## 2 Related Work

Our audit examines a cited rationale at four levels: the decision, the citations, the claims made about the cited evidence, and the reliance of the decision on that evidence. We relate each level to prior work, then turn to LLMs as decision makers and to risk control.

**Citation quality and attribution.** Retrieval-augmented generation conditions outputs on retrieved documents (Lewis et al., 2020), and attribution research asks whether the resulting text is supported by what it cites. AIS defines attribution to identified sources (Rashkin et al., 2023); Attributed QA and ALCE benchmark attribution and citation quality (Bohnet et al., 2022; Gao et al., 2023b); RARR revises text to restore attribution (Gao et al., 2023a); and generative search engines leave many statements unsupported by the sources they cite (Liu et al., 2023). These studies ask whether each citation supports its statement. We additionally ask which evidence gets cited at all, and find citations that are valid yet one-sided and concentrated at the start of the evidence list (§5.2).

**Claim-level accuracy against data.** FActScore checks atomic facts against a knowledge source (Min et al., 2023). Data-to-text generation has long documented factual errors in text produced from tables (Wiseman et al., 2017); Thomson and Reiter (2020) annotate every error in generated basketball summaries against the source data, including incorrect numbers, names and words, and summarization work distinguishes intrinsic from extrinsic hallucination (Maynez et al., 2020; Ji et al., 2023). Statements requiring comparison, superlatives or aggregation over a table are hard both to verify and to generate faithfully (Chen et al., 2020b,a). We apply this error-annotation practice to the free-text rationales

of instruction-tuned LLMs, where the supplied evidence is small enough for every claim to be checked exactly, and find the same superlative, range and quantifier errors (§5.3).

**Faithfulness and reliance.** Faithfulness must be evaluated separately from plausibility (Jacovi and Goldberg, 2020). ERASER measures the comprehensiveness and sufficiency of rationales by removing them (DeYoung et al., 2020); counterfactual and reconstruction tests probe natural-language explanations (Atanasova et al., 2023); and chain-of-thought explanations can omit the factors that drive a prediction (Turpin et al., 2023) or be tested by intervening on the reasoning (Lanham et al., 2023). Self-explanation faithfulness varies across models and tasks (Madsen et al., 2024), and many such tests measure output-level self-consistency rather than faithfulness to internal computation (Parcalabescu and Frank, 2024); ContextCite attributes generations to context by ablation (Cohen-Wang et al., 2024). Our matched deletion is an ERASER-style comprehensiveness test with a same-kind uncited control and, in the terms of Parcalabescu and Frank (2024), a self-consistency test (§5.4).

**How models use supplied context.** Models may ignore context that conflicts with parametric knowledge (Longpre et al., 2021) or accept coherent counter-evidence while favoring their priors (Xie et al., 2024); irrelevant context distracts them (Shi et al., 2023), and information in the middle of long contexts is used less (Liu et al., 2024). Our evidence ablation examines the complementary case in which the context is decisive and the prior is misleading (§5.1), and our citation analysis adds a position effect in what models choose to cite.

**LLMs as decision makers.** LLMs can optimize by proposing candidates from past evaluations (Yang et al., 2024), and configuration advisors mine tuning hints from manuals (Trummer, 2022), prune Bayesian-optimization searches with LLM-extracted knowledge (Lao et al., 2024), generate complete configurations (Giannakouris and Trummer, 2025) or tune Linux kernel build options as agents (Chen et al., 2024; Lin et al., 2025); classical tuners use Gaussian processes, reinforcement learning or black-box optimization (Van Aken et al., 2017; Zhang et al., 2019; Herodotou et al., 2021; Snoek et al., 2012; Golovin et al., 2017; Lee et al., 2022). These works score the chosen action end to end, whereas we ask what the model adds over a selector given the same evidence and whether its stated reasons are correct; we do not compare with these systems head to head. Following arguments against random or standard splits (Gorman and Bedrick, 2019; Søgaard et al., 2021) and for reporting matched budgets (Dodge et al., 2019), our data roles come from disjoint operating conditions and every baseline has an explicit budget.

**Uncertainty and risk control.** Self-consistency aggregates sampled outputs (Wang et al., 2023); LLMs can partly predict their own correctness (Kadavath et al., 2022), verbalized confidence can be better calibrated than token probabilities for RLHF models (Tian et al., 2023) yet remains overconfident (Xiong et al., 2024), and semantic entropy measures uncertainty over meanings (Kuhn et al., 2023). Selective classification abstains to meet a target risk (Geifman and El-Yaniv, 2017), and conformal methods give guarantees for language-model outputs (Quach et al., 2024), their factuality (Mohri and Hashimoto, 2024) and monotone losses (Angelopoulos et al., 2024), with extensions for non-exchangeable or drifting data (Farinhas et al., 2024; Barber et al., 2023; Gibbs and Candès, 2021); none implies that a window reset after a detected change (Bifet and Gavaldà, 2007) restores a fixed risk level. Our rank gate is a restricted class-conditional instance of this family, and we report where its assumptions fail (§5.5).

## 3 Testbed: GroundKern

GroundKern is an evidence-grounded generation pipeline for system configuration that we use as an instrument (Figure 2). Four properties matter for the audit. The evidence is *addressable* (every citation resolves to one item), *portable* (a non-LLM selector can receive the same object) and *editable* (kinds or items can be removed), and enforcement is *independent* of generation, so its errors can be counted against labels the generator never sees.

**Evidence as a structured context.** The store holds typed items, each with a stable ID, a kind and a provenance record: summary tables, individual measurement traces, and edges of a knob-interaction graph. The graph is a small knowledge representation with a strict admission rule: for a lower-is-better loss $L$ and baseline values $a_0, b_0$, an edge requires the contrast $I_{ab} = L(a,b) - L(a,b_0) - L(a_0,b) + L(a_0,b_0)$ from independently measured configurations, with a bootstrap interval

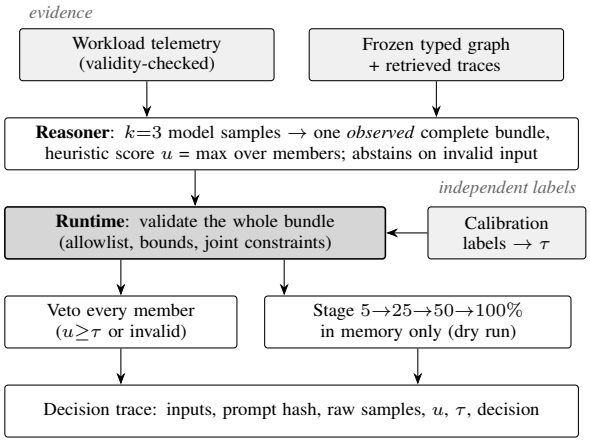



Figure 2: GroundKern decision path. Evidence, generation and enforcement are separate components: the frozen evidence object can be given to a non-LLM selector, ablated or partially deleted, citations can be resolved against its item IDs, and gate errors can be counted against independently defined labels. The REST runtime is a dry run; a separate per-thread actuator applies, verifies and restores timer slack and affinity in owned child processes.

over runs that excludes zero, and records its scope, values, measurement IDs and interval. Edges describe empirical interactions, not causal couplings (Appendix A). For the audit, the evidence is frozen and serialized as a JSON list that is identical for every advisor.

**Generation.** The reasoner retrieves typed neighbors and lexically matched trace text (Lewis et al., 2020) and asks a model for a complete proposal, a set of knob values, with cited item IDs and an explanation. In deployment it draws $k = 3$ samples (Wang et al., 2023) and returns a modal complete proposal actually present in a response, rather than splicing per-knob winners into an unseen combination, scores each member by sample disagreement $u$ (a heuristic, not a calibrated probability; Appendix A), and abstains when no model responds or the context budget would force evidence to be truncated. The audit in §4 calls each model directly on the frozen evidence, one response per seed, so that decisions, citations and explanations can be compared across models without the aggregation step.

**Enforcement.** The runtime treats a proposal as a unit: it validates every member (allowlisted knobs, bounds, joint constraints, no duplicates) and, if any member score satisfies $u \geq \tau$, vetoes all of them; otherwise the proposal enters a staged rollout. The

Table 1: Audit levels, in the order of §5; all use the same frozen evidence.

| Level | Manipulation | Measure |
|---|---|---|
| Decision | matched non-LLM selector; evidence ablation | held-out P95 change vs. control |
| Citation | — (observational) | kind, configuration, position |
| Claim | — (observational) | labels against the evidence |
| Reliance | cited vs. matched uncited deletion | decision change |

REST runtime is a dry run that never writes kernel settings (Appendix A). The threshold $\tau$ comes from a rank gate, a selective-prediction rule (Geifman and El-Yaniv, 2017) calibrated on unsafe examples only, where $Y = 1$ marks an unsafe outcome defined from raw observations, never from whether a veto or rollback occurred. If unsafe calibration and test scores are exchangeable, the gate accepts an unsafe action with probability at most $\alpha$ (Proposition 1, Appendix A); this bounds misses among unsafe actions, not the unsafe fraction among accepted ones. The proof, an expected-cost identity, a discussion of drift and the bounded per-thread actuator used in §5.5 are in Appendix A.

## 4 Audit Protocol

The protocol treats configuration advice as a grounded generation task and audits its output at four levels, each with its own manipulation and measure (Table 1).

**Task.** The input is a list of four allowed configuration labels and a JSON evidence list; the output is one label, up to six cited evidence IDs and a free-text explanation. The labels name per-thread settings: timer slack of 50 $\mu$s or 1 ms crossed with affinity to all four CPUs or to CPU 0 (`s50000_aall`, `s50000_aone`, `s1000000_aall`, `s1000000_aone`). They are opaque by design: nothing in a label says which setting is better. The objective is the 95th-percentile (P95) wake-up lateness; `s50000_aall` is the control. Runs execute in fresh child processes on one Linux host (Appendix B).

**Data roles and frozen evidence.** Training, retrieval, calibration and test data come from disjoint wake-up periods of 2, 3, 4 and $\{5, 8\}$ ms (Søgaard et al., 2021). Each period has ten randomized blocks of the four configurations with 40 recorded events per run: 200 runs and 8,000 events, collected

under a plan saved in advance. The evidence contains three kinds of items: the training table (mean training-run P95 per configuration), an interaction-graph item (the training contrast, $-18.6\,\mu$s, whose prespecified interval includes zero, so the graph has no edge) and 40 retrieval items, one per retrieval run, listed in block order. Selections and thresholds are frozen, with hashes, before calibration and test data are opened; test outcomes never enter a prompt.

**Decisions.** The equally informed baseline is a table selector that picks the minimum of the training table; a saturated two-factor model of the same means is equivalent. It uses the same training runs as the LLMs and no online trials; matched-budget bandit, RL and BO optimizers (Auer et al., 2002; Snoek et al., 2012) are further references (Appendix B). Ten LLMs receive the same system prompt, evidence and output schema (Appendix C) at temperature 0 with five seeds: Qwen2.5 7B and 14B (Qwen Team, 2024), Llama-3.1-8B (Grattafiori et al., 2024) and Phi-4 (Abdin et al., 2024) locally as hashed 4-bit GGUF files in llama.cpp[1], and Llama-3.3-70B, Qwen3-235B-A22B (Yang et al., 2025), DeepSeek-V3.2 (DeepSeek-AI, 2025), Nemotron-3-Ultra-550B (NVIDIA, 2026), Gemini 3.8 Flash (Google DeepMind, 2026b) and Gemini 3.1 Pro preview (Google DeepMind, 2026a) through OpenRouter[2], whose weights cannot be hashed. Four input variants isolate the evidence: full, no graph, no retrieval, and no evidence. Each choice is scored by paired replay on the held-out grid: its run P95 in each test block and its change relative to the control in the same block, with a Student-$t$ interval over ten blocks.

**Citations.** For every valid full-context answer we record the kind of each cited item, whether a cited run belongs to the chosen or to a competing configuration, and its position in the evidence list. Since the schema admits only valid IDs, these counts describe which evidence a model presents.

**Claims.** Following error annotation in data-to-text evaluation (Thomson and Reiter, 2020), every quantitative or comparative claim in each distinct explanation is checked against the prompt evidence and labeled *correct*, *overstated*, *wrong* (including partly wrong), *wrong unit*, *wrong attribution* (a value assigned to the wrong source) or, for no-evidence prompts, *unsupported prior*. A claim is *misstated* if it is wrong, misattributed, overstated or in the wrong unit; the single partly wrong claim is reported but not counted, which makes the counts conservative. A list-valued statement counts as one claim and a comparison as another; "consistently" or "always" lowest is overstated, since the chosen configuration is lowest in 8 of 10 retrieval blocks. Local models returned byte-identical text across seeds, so each distinct explanation is audited once per variant; hosted models varied across seeds and providers, so all distinct full-context explanations are audited. DeepSeek-V3.2's providers ignored the 400-character limit in 14 of 20 answers; these are audited in full. An AI assistant drafted the labels under these rules and one author verified all, changing none; two blind non-author annotators agreed ($\kappa = 0.92, 0.86$), and the three-way majority equals every verified label (App. C).

**Reliance.** Following the comprehensiveness test of ERASER (DeYoung et al., 2020), for each seed we delete the items cited in the full-context answer and, separately, the same number of uncited items of the same kinds. A decision that changes only under cited deletion indicates citation-specific reliance. A seed is non-estimable when no same-kind control exists, as for the training table, the only item of its kind.

## 5 Results

We report the audit at four levels, from the decision to the text that justifies it: what models choose (§5.1), what they cite (§5.2), what their explanations claim (§5.3) and whether the cited evidence matters (§5.4). A final subsection evaluates the testbed's risk gate (§5.5).

### 5.1 Decisions: Evidence Is Necessary and Sufficient

The table selector and its saturated-model equivalent choose the control, which is also best on the test grid, so any other choice is a loss; control test P95 is $179.7 \pm 102.0$ and $155.7 \pm 64.4\,\mu$s at 5 and 8 ms (95% half-widths across ten blocks). Matched-budget random search, Q-learning, UCB1 and BO converge on it as well, choosing it in 98–99% of 1,000 seeds at $B = 40$ (Appendix B).

Table 2 gives each model's choice under every input condition. With evidence, nine of ten mod-

[1] https://github.com/ggml-org/llama.cpp, via llama-cpp-python 0.3.35.

[2] https://openrouter.ai

Table 2: LLM decisions with evidence (full, no graph, no retrieval), without it, and after deleting cited or matched uncited evidence; superscripts count seeds; n/e: non-estimable; †hosted. Paired test P95 change versus control at 5 and 8 ms (%, 95% half-width): ctrl 0; 50/1 $-145 \pm 125$ and $-180 \pm 156$; 1m/all $-599 \pm 188$ and $-631 \pm 179$; 1m/1 $-631 \pm 183$ and $-699 \pm 206$.

| Model | Evidence | None | Del. cited | Del. unc. |
|---|---|---|---|---|
| Qwen-7B | ctrl | 50/1 | n/e | n/e |
| Llama-8B | 50/1 | 50/1 | 1m/all | ctrl |
| Qwen-14B | ctrl | 50/1 | n/e | n/e |
| Phi-4 | ctrl | 1m/1 | 50/1 | ctrl |
| Llama-70B† | ctrl | 1m/1 | ctrl | ctrl |
| Qwen3-235B† | ctrl | 1m/1[3] ctrl 50/1 | ctrl | ctrl |
| DeepSeek-V3.2† | ctrl | 50/1 | n/e | n/e |
| Nemotron-550B† | ctrl | ctrl | ctrl | ctrl |
| Gemini-Flash† | ctrl | 50/1 | n/e | n/e |
| Gemini-Pro† | ctrl | 50/1 | n/e | n/e |

els choose the control in all three evidence variants, exactly as table lookup does; Llama-3.1-8B instead picks `s50000_aone`, which is worse on replay ($-145 \pm 125\%$ and $-180 \pm 156\%$). Removing the graph item changes no choice, and removing the retrieval runs changes none either: the training table alone suffices. Without evidence, every model except Nemotron, which declares the default, chooses a worse setting in at least four of five seeds, usually single-CPU affinity with 50 $\mu$s slack (six models) or with 1 ms slack (three). The evidence is therefore necessary; in this task it is also sufficient, since no model improves on the selector that simply reads it. 249 of 250 calls returned schema-valid output.

### 5.2 Citations: Valid, One-Sided and Front-Loaded

Every citation refers to a supplied item, since the schema permits nothing else, but what models choose to cite is far from a balanced account of the evidence. Across the 50 full-context answers, 285 citations comprise 25 of the training table, none of the interaction-graph item, and 260 of individual retrieval runs. Of these 260, 252 are runs of the configuration the model chose; only Gemini 3.1 Pro ever cites a run of a competing configuration. The cited evidence is thus confirmatory: no answer from the other nine models cites a run of a competing configuration, although `s50000_aone` beats the control in two of ten retrieval blocks.

Citations also favor the start of the evidence list. Retrieval runs appear in block order, and 209 of the 260 retrieval citations fall in the first five of ten blocks. Seven models cite exactly a prefix of the blocks (typically 0–4 or 0–5) and so never cite the chosen configuration's lowest run. Rerunning the hosted models (Appendix E) shows that, under the cap, the prefix follows list position (Liu et al., 2024): with the runs reversed, four of six models cite those listed first; uncapped, models cite 9.0 runs, all of the chosen configuration.

Table 3: Verification of quantitative and comparative claims in explanations against the supplied evidence (AI-drafted labels, verified by an author). Corr.: correct; Over.: overstated; Wrong: wrong or partly wrong value; Unit/Src: wrong unit or evidence source; Prior: unsupported prior without evidence; Misst.: wrong, misattributed, overstated or wrong unit (the one partly wrong claim is not counted). Generated by `claim_taxonomy.py`.

| Model | $n$ | Corr. | Over. | Wrong | Unit/Src | Prior | Misst. |
|---|---|---|---|---|---|---|---|
| Qwen-7B | 5 | 1 | 1 | 1 | 1 | 1 | 3 |
| Llama-8B | 7 | – | – | 6 | – | 1 | 5 |
| Qwen-14B | 6 | 1 | 1 | – | 3 | 1 | 4 |
| Phi-4 | 10 | 5 | 3 | 1 | – | 1 | 4 |
| *Local, all variants* | *28* | *7* | *5* | *8* | *4* | *4* | *16* |
| Llama-70B | 7 | 5 | – | 2 | – | – | 2 |
| Qwen3-235B | 6 | 3 | 3 | – | – | – | 3 |
| DeepSeek-V3.2 | 16 | 10 | 2 | 4 | – | – | 6 |
| Nemotron-550B | 23 | 13 | 5 | 5 | – | – | 10 |
| Gemini-Flash | 12 | 10 | 2 | – | – | – | 2 |
| Gemini-Pro | 5 | 3 | 2 | – | – | – | 2 |
| *Hosted, full context* | *69* | *44* | *14* | *11* | – | – | *25* |

Table 4: Examples of misstated claims (paraphrased as audited) and the check against the supplied evidence; all latencies in $\mu$s.

| Model, label | Claim | Evidence |
|---|---|---|
| Gemini-Pro†, overstated | control "consistently the lowest in retrieval runs" | lowest in 8 of 10 blocks |
| Nemotron†, wrong | `s50000_aone` ranges from 85.0 to 430.7 | maximum is 571.84 |
| DeepSeek†, wrong | control "consistently lower than any" `s50000_aone` run | ranges overlap: $85.033 < 183.217$ |
| Qwen-7B, wrong source | training table gives control 126.36 | 126.36 is one retrieval run; training mean 132.94 |
| Qwen-14B, wrong unit | training table gives control 132.9413 ms | evidence is in $\mu$s |
| Qwen-14B, prior | "50000" denotes a smaller dataset size | no evidence given; name is opaque |

### 5.3 Explanations: Cited but Not Correct

The explanations are often wrong about what the cited items say (Table 3): 16 of 28 audited claims from local models and 25 of 69 from hosted models are misstated; with either blind annotator's labels instead, 37 or 38 of 97 are. Errors fall into four recurring patterns (Table 4).

*Overstated regularities.* Thirteen of the 19 overstatements assert that the chosen configuration is lowest "consistently" or in "all" retrieval runs, whereas it is lowest in 8 of 10 retrieval blocks; the other six claim "significant" differences that no test in the evidence supports. These claims are easier to make when, as above, the cited runs are all of one configuration.

*Range and extremum errors.* All 11 wrong values from hosted models misreport a range, an extremum or a cross-configuration ordering. Nemotron repeatedly gives the `s50000_aone` range as 85.0–430.7 $\mu$s (its maximum is 571.84), and DeepSeek-V3.2 states that the control is "consistently lower than any" `s50000_aone` run although the ranges overlap (Figure 1). Llama-3.1-8B reports 571.84 $\mu$s, the *maximum* of `s50000_aone`, as the lowest P95, consistent with its wrong choice.

*Source and unit confusion.* Qwen2.5-7B attributes the retrieval value 126.36 to the training table, whose mean is 132.94; Qwen2.5-14B reports the training mean 132.9413 in milliseconds in three explanations, although the evidence declares microseconds.

*Invented semantics.* Without evidence, all four local models justify their choice by assigning meanings to opaque identifiers, for instance reading "50000" as a smaller dataset size or "one" as a more aggressive approach. The configuration names carry no such information.

These errors are largely decoupled from the decision: 35 of the 41 misstated claims accompany a choice of the control. Nor does variation in wording track variation in decisions: hosted models produced two to five distinct full-context explanations each while choosing the control in every seed. Valid citations thus establish provenance, and a correct decision can coexist with an incorrect account of the evidence that supports it.

### 5.4 Reliance: Rarely Estimable, Rarely Specific

Five models cite the training table in their full-context answers, so matched deletion is non-estimable for them. Among estimable models, only Phi-4 shows citation-specific reliance: deleting its six cited retrieval runs changes its choice in all five seeds while matched uncited deletion does not, but the new choice (`s50000_aone`) contradicts the training table that remains in the prompt. Llama-3.1-8B changes its choice under both deletions, which indicates sensitivity to context rather than reliance on what it cites. Llama-3.3-70B, Qwen3 and Nemotron change under neither, which is consistent with the remaining evidence still favoring the control but uninformative about reliance.

### 5.5 Enforcement: A Calibrated Gate under Shift

A test run is *unsafe* when more than 10% of its 40 events exceed a predeclared 200 $\mu$s deadline, a label computed from timing alone. For the gate, the score is a configuration's training deadline-miss fraction, fixed before calibration on 4 ms runs only; all three prespecified $\alpha \in \{.05, .10, .20\}$ yield $\tau = .9225$ because the unsafe-score order statistics are tied. Every candidate is executed in an isolated process, so accepted and vetoed candidates both have observed outcomes (Figure 3 in Appendix B).

Unsafe acceptance is $5/25 = 20.0\%$ and $4/24 = 16.7\%$ among unsafe runs, significantly above the 5% level for this calibration draw (one-sided binomial $p = .007$ and .03) but within sampling error of 10% ($p = .10, .21$); unsafe fractions among accepted runs are $5/20$ and $4/20$, and veto precision is $20/20$ in each period. As with LLM explanations, a nominally calibrated signal is only as good as the exchangeability it assumes, and a period shift can be enough to break it. The bounded actuator restored both original settings in all 80 staged executions (Appendix B).

## 6 Discussion

**Decisions and rationales come apart.** Most misstatements accompany a correct choice. This pattern is consistent with a decision driven by a simple reading of the evidence, here the minimum of the training table, and a rationale that is generated as a separate summary of the retrieval runs. Choosing the smallest of four means is easy; summarizing forty noisy measurements with ranges, extremes and quantifiers is where the errors concentrate. Because the decision is right, an end-to-end evaluation would never surface these errors, and a reader who trusts the rationale could come away with a false picture of the evidence, for instance that the chosen configuration is never beaten. This mirrors findings that explanations can be plausible yet unfaithful to what drives a prediction (Jacovi and Goldberg, 2020; Turpin et al., 2023), with the difference that here the error is checkable against the prompt itself.

**Citations as selective presentation.** A citation list is also an editorial choice. Nine of ten models

cited only runs of the configuration they chose, and most cited the first runs in the list, so a reader who follows the citations sees supporting evidence only and never the blocks in which a competitor wins. It also licenses overstatement (§5.3): "consistently the lowest" cannot be contradicted by the cited runs, only by those left uncited. The six-citation cap explains the preference for early items, which follows list position, but not the preference for confirming ones, which persists without it (Appendix E).

**Scale reduces but does not remove misstatement.** Every audited model misstated at least two claims, and "consistently the lowest" recurs across families. On full-context prompts, misstatement falls with size among open-weight models (7/8 claims at 7–15B, 21/52 at 70–671B; Spearman $\rho = -0.78$ over eight models, a trend on few claims; Appendix D) yet stays at 17–50% per hosted model; unit and source errors occur only at small scale, overstatements and wrong values at every scale. Only Gemini-Pro cites other configurations. The superlative, range and quantifier errors long documented in data-to-text generation (Thomson and Reiter, 2020; Chen et al., 2020a) are therefore not removed by instruction tuning or scale in the models we tested.

**Implications for attribution metrics.** Attribution benchmarks ask whether each cited passage supports its statement, often with an entailment judge (Rashkin et al., 2023; Gao et al., 2023b). One-sided citation lists are invisible to such per-citation scores, and many misstated claims quote correct numbers and err only in a comparison or quantifier, near-misses that eight judges, including the fact-checkers AlignScore and MiniCheck, fail to separate from correct claims (balanced accuracy 0.47–0.68; Appendix C). Evaluations should therefore also measure what is left out, such as the share of contrary evidence cited, and check claims exactly where possible.

**What deletion can and cannot show.** Matched deletion measures whether a decision is sensitive to what is cited, which is a form of self-consistency rather than faithfulness to internal computation (Parcalabescu and Frank, 2024). Even that is estimable only when the evidence contains redundant items of the same kind: half of our models cited the single training table, for which no control exists. Evidence sets for such audits should be designed with matched alternatives for every item a model might cite.

**When could an LLM add value?** Our evidence determines the best choice, so a selector that reads it is hard to beat. A language model could add value when the evidence is incomplete and prior knowledge, for example from documentation, carries information the measurements lack. Testing this requires tasks in which the default is not optimal and in which the baseline receives the same documents; the invented label meanings in our no-evidence runs show why such priors must be checked, not assumed.

## 7 Conclusion

A rationale that cites evidence promises a good decision, true claims about the evidence, and reliance on what it cites. Auditing them separately for LLM configuration advice, we found that they came apart. Ten LLMs needed the evidence to choose well but added nothing beyond a selector that reads it; their citations were valid but one-sided (252 of 260 cited runs back the chosen configuration); 41 of 97 audited claims misstated validly cited evidence, mostly alongside a correct decision; and deletion showed citation-specific reliance for only one model.

For NLP evaluation, these results suggest three practices. First, citation validity and passage-level support are not enough: an explanation can cite valid items, omit the contrary ones and still misreport what it cites, so attribution should be checked claim by claim and for what is left out. Second, the errors that data-to-text research documented for tables, namely wrong numbers, superlatives, ranges and overstated regularities, persist in the free-text rationales of instruction-tuned models of every size we tested, and deserve targeted checks. Third, LLM decision support should be compared with a baseline given the same evidence, reporting decision quality and rationale correctness separately. Next steps are tasks where documentation priors could help, and stronger automatic claim checks.

## Limitations

**Task and generality.** The audit covers one grounded-advice task with four candidate labels, English prompts and a small numeric evidence set that determines the best choice, and the default is best on the test grid. It can therefore expose missing

added value and misreported evidence, but it cannot show that an LLM adds value on harder tasks, and its error rates need not transfer to domains with textual evidence (manuals, logs, documentation), longer contexts or other languages. The configuration labels are opaque by design; with descriptive labels, models' priors could help or hurt differently.

**Annotation.** Claim labels were drafted with an AI assistant and verified by one author; agreement with two independent blind annotators who are not authors is high (Cohen's $\kappa = 0.92$ and $0.86$; Fleiss' $\kappa = 0.85$ with the verified labels), but rests on few annotators and a small sample (97 claims). The annotators labeled claims that had already been extracted, so the extraction step (which statements count as claims) was not independently validated, and only quantitative or comparative claims were audited. Segmenting explanations into claims and deciding when a claim is "overstated" involve judgment; we fixed both rules in advance (a list counts as one claim, a comparison as another; "consistently" is overstated when a configuration is lowest in 8 of 10 blocks), and supply every label with its check so that the annotation can be repeated or automated.

**Models and prompting.** The main audit uses one system prompt and output schema at temperature 0. A rerun of the hosted models with a paraphrased prompt, reordered evidence, no citation cap and sampling (Appendix E) leaves decisions and one-sided citation unchanged, but scores its explanations only with automatic proxies, not verified labels; local models, evidence formatting and reasoning modes were not varied. Local explanations were byte-identical across seeds, and five greedy seeds are not independent samples. Local models were 4-bit quantized and may differ from full-precision weights. Hosted models were served by several providers, could not be hashed, varied across seeds and are not reproducible bit for bit; DeepSeek-V3.2's providers ignored the explanation length limit. The six-citation cap and the 400-character limit shape what models cite and write; the robustness rerun shows that the citation prefix depends on the cap.

**Reliance.** Matched deletion measures the sensitivity of a decision to cited evidence, a form of self-consistency, not faithfulness to internal computation. It was non-estimable for half of the models, because they cited the only item of its kind.

**Measurement scope.** All measurements use one periodic workload family on one shared host, with data roles separated by period rather than by workload family or hardware; run-level intervals do not account for uncontrolled interference or correlated system state and are unadjusted for multiple comparisons. Measured graph induction emits no edge, so the graph item is uninformative in this task. Selector, BO and RL comparisons replay recorded runs rather than executing closed loops, and because control is best they can reveal only losses. The telemetry prototype's scheduler and block-I/O proxies are not end-to-end request latencies, and the local benchmarks in the appendix do not substitute for application workloads such as web serving, databases or GPU inference.

**System and safety scope.** The REST runtime is a dry run that never writes kernel settings, and the separate actuator changes only an owned child thread's timer slack and affinity; neither establishes traffic isolation, a deployed multi-knob actuator or recovery from irreversible effects. The reasoner awaits model samples synchronously, and its logs are append-only files without tamper resistance. Prompting a model to respect graph edges does not enforce them; only the runtime's explicit constraints are enforced. The rank gate's assumptions fail under drift, selective labeling or a score updated with calibration data, as the period-shift result shows, and the gate replay neither verifies deployment assumptions nor establishes recovery. We therefore make no LLM-superiority, deployment-efficacy, no-worse-than-default or production-safety claim; Section 4 and Appendix A describe the comparisons and engineering such claims would require.

## Ethics Statement

**Intended use and risks of the findings.** The paper studies how far users can trust LLM rationales that cite supplied evidence. Its main risk is misreading in either direction: treating valid citations or a correct final choice as evidence that a rationale is accurate, which our results argue against, or reading the audit as evidence that LLM configuration advisors are safe or effective, which it does not show (see Limitations). We recommend that LLM advice affecting real systems be paired with claim-level checks of its rationale and with enforcement that does not rely on the model.

**Data, annotation and models.** The data are timing measurements of local software workloads, synthetic configuration labels, and model responses to one fixed technical prompt; they contain no personal data and involve no human participants. The explanation-claim labels were drafted with an AI assistant and verified by an author (see Use of AI assistants); two further annotators, fellow researchers who are not authors, volunteered without payment to label them blind and independently for an agreement study and consented to the release of the labels. All models were used for research inference under their licenses: Apache 2.0 (Qwen), the Llama 3.1 and 3.3 Community Licenses, MIT (Phi-4, DeepSeek-V3.2), OpenMDW-1.1 (Nemotron) and the Gemini API terms. Code, measurements, prompts, responses and annotations are provided as supplementary material and will be released publicly under Apache 2.0; the stored responses remain subject to each model's terms.

**System safety.** The REST runtime performs no kernel writes; the experimental adapter changes only its owned child thread's timer slack and affinity, restores both settings, and performs no host-wide writes. Broader interventions require an isolated testbed, documented rollback limits and appropriate authorization.

**Compute.** The study used about 3.2 CPU-hours of local inference and CPU-only attribution judges without a GPU, 300 hosted API calls (150 in the main audit, about 0.5 USD; 150 in the robustness rerun, 0.83 USD), and seconds of kernel measurement (Appendix C).

**Use of AI assistants.** The research idea, problem formulation and study design are the authors'. AI assistants were used for writing and debugging code, for drafting and restructuring prose, for locating references and for drafting the explanation-claim labels, which an author then verified; every reference was checked against its publisher, ACL Anthology, Crossref, proceedings or arXiv record, and every claim label is released with its check so that it can be verified. The authors take full responsibility for the content. Every reported number is produced by the supplied scripts from raw logs whose SHA-256 hashes are recorded, so that it can be recomputed independently; no generated value substitutes for an unavailable observation.

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## A GroundKern Testbed Details

**Design hypotheses.** GroundKern is motivated by three hypotheses. First, a configuration's utility may depend on workload and hardware. Second, joint knob effects may be non-additive. Third, performance, failure probability and intervention cost may conflict. Each hypothesis requires measured interventions with explicit baselines.

For a loss $L$, define an interaction contrast from four independently measured configurations:

$$I_{ab} = L(a, b) - L(a, b_0) - L(a_0, b) + L(a_0, b_0).$$

For lower-is-better loss, negative $I_{ab}$ indicates a beneficial non-additive interaction relative to the chosen baseline. A bootstrap interval must resample independent runs, not individual requests from a single run. The sign can change with workload, value range or kernel version, so an edge needs its scope, configuration values, measurement IDs and uncertainty interval. Such an edge describes an empirical interaction, not an unconditional causal coupling.

A DEPENDS-ON edge requires a validated admissibility rule or measured range conditioning; it cannot be inferred solely from the sign of $I_{ab}$. The seed graph is empty until edges pass this test. The principal question is whether a reasoner can use independently acquired evidence more effectively than an optimizer with the same inputs, after charging the cost of acquiring that evidence.

**Restricted rank gate.** The gate of §3 sets $\tau$ as follows; $Y = 1$ marks an unsafe outcome.

**Proposition 1** (Restricted rank gate)**.** *Fix a score function before calibration. Suppose $n$ unsafe calibration scores and the next unsafe test score are jointly exchangeable. Set $j = \lfloor \alpha(n+1) \rfloor$ for $0 < \alpha < 1$. If $j = 0$, veto all; otherwise let $\tau$ be the $j$th smallest unsafe calibration score and accept only when $u < \tau$. Then the unsafe-class acceptance probability, marginal over calibration and test scores, is at most $j/(n+1) \le \alpha$.*

*Proof.* By exchangeability the test score's rank among the $n + 1$ scores is uniform; it falls below the $j$th calibration score only in the first $j$ ranks, and ties only reduce this. □

The bound controls misses among unsafe actions, $\Pr(A = 1 \mid Y = 1)$, not the unsafe fraction among accepted actions; a cost-selected threshold is capped at the rank threshold, and no unsafe calibration example implies veto-all. The gate accepts any fixed score; §5.5 evaluates it with a measured risk score rather than $u$.

**Disagreement score.** For a knob–value pair proposed in $v$ of $k$ samples, the reasoner's score is $u = 0.7(1-v/k)+0.3(1-w)$, clipped to $[0.02, 1]$, where $w$ is the strongest retrieved synergy weight (zero when absent). It is neither predictive entropy nor a calibrated probability. Failed samples stay in the denominator, and repeated copies of a fallback are not counted as independent votes.

**Algorithm 1** Implemented dry-run bundle path

**Require:** Validated bundle $B$, fixed score function, calibration threshold $\tau$

Validate every member and joint constraints before changing state
**if** another bundle is active **then**
  Return conflict
**end if**
**if** $\max_{a \in B} u(a) \ge \tau$ **then**
  Record veto for all members; return
**end if**
Stage the complete bundle in memory at 5%
**for** requested stage in $\{25, 50, 100\}\%$ **do**
  Advance the simulated bundle; record the transition
**end for**
On cancellation, clear the complete simulated bundle
Never infer an unsafe/safe training label from simulated completion

**Bundle contract.** A request contains a nonempty bundle identifier and a list of recommendations with `id`, `knob`, `proposed`, `uncertainty`, and optional rationale and provenance. Each request submits one bundle. Duplicate knobs, nonfinite scores, unsupported controls, and example-policy bound violations reject the entire request. Dirty-ratio changes must supply both values and satisfy the example policy that the background ratio is less than the foreground ratio. This is a declared policy, not proof that a configuration improves any workload. If any score satisfies $u \ge \tau$, all members are vetoed; otherwise all are staged together. A concurrent bundle receives a conflict response. Dry-run results return `applied=[]` and list simulated members separately.

**Deployment requirements.** A real actuator must snapshot all old values, validate the full target configuration and available host interfaces, serialize updates, verify readback, and restore every old value on failure. Multiple sysctl writes are not a kernel transaction: partial exposure must be measured and cannot be described as atomic hardware state. Canary shares require separate hosts or an appropriate isolation mechanism; changing a host-global sysctl cannot isolate a fraction of requests on that host. Missing, stale or malformed end-to-end

telemetry must stop promotion. These capabilities are requirements, not features demonstrated by the dry-run state machine.

**One illustrative decision trace.** Consider supplied context with example values 5 and 15 for the two dirty ratios, and retrieved evidence IDs $e_1, t_1$. Suppose all three model samples agree on this pair and $w = 0$, giving $u = 0.30$. If an independently obtained threshold is $\tau = 0.25$, the whole bundle is vetoed; if $\tau = 0.35$, it can enter the dry-run canary after joint validation. The audit record includes the values, scores, threshold and decision. These numbers illustrate the contract and are not a measured recommendation or calibrated operating point. The initial artifact threshold is zero until calibration data is explicitly supplied.

**Actuator.** A separate adapter snapshots a child thread's timer slack and CPU affinity (Linux man-pages project, 2026a,b), validates the target, applies both settings, verifies readback and restores both originals in a `finally` path, including after partial application. Updates are sequential, not an atomic kernel transaction.

**Expected cost of gated acceptance.** Let $Y = 1$ denote an independently defined unsafe outcome and $A = 1$ acceptance. An outcome label is not defined by whether a rollback happened. For example, a benchmark can predeclare a request deadline, count observed failures or deadline exceedances, and define an action-level breach from these raw observations in a fixed window. Rejected actions have unknown counterfactual outcomes unless evaluated in an isolated experiment; they must not automatically be labeled safe or unsafe.

**Theorem 1** (Cost identity)**.** *For costs $c_{\mathrm{FN}}, c_{\mathrm{FP}}, \lambda \geq 0$, define $C = c_{\mathrm{FN}}\mathbf{1}\{Y = 1, A = 1\} + c_{\mathrm{FP}}\mathbf{1}\{Y = 0, A = 0\} + \lambda D$, where $D \geq 0$ is measured exposure or SLO debt. Write $\pi = \Pr(Y = 1)$, $q = \Pr(A = 1 \mid Y = 1)$ and $r = \Pr(A = 0 \mid Y = 0)$. Then*

$$\mathbb{E}C = c_{\mathrm{FN}}\pi q + c_{\mathrm{FP}}(1 - \pi)r + \lambda \mathbb{E}D.$$

*For a zero-probability class its associated product is zero.*

*Proof.* Expand each indicator expectation as its joint probability and use the definition of conditional probability. Linearity requires no independence. □

This is an accounting identity, not a claim of novelty or a safety bound. A bound $q \leq \alpha$ controls misses among unsafe actions. A different bound $\Pr(Y = 1 \mid A = 1) \leq \alpha$ instead gives $\Pr(Y = 1, A = 1) \leq \alpha \Pr(A = 1)$, not $\pi\alpha$. Likewise $r$ is a veto rate among safe actions, not the proportion of vetoed actions that are safe. We report all denominators separately.

For bundle-level calibration, $u$ denotes the maximum member score and $Y$ the independently observed bundle outcome. Per-knob labels alone do not establish a bundle-level guarantee.

Selecting a smaller threshold by a cost rule cannot enlarge the acceptance event. Selecting a larger threshold can invalidate the bound; therefore the implementation uses the minimum of the rank threshold and optional cost-selected cap. No unsafe calibration examples implies veto-all. Calibration-window diagnostics are not test-set estimates.

**Drift and selective observation.** Sliding windows and a drift detector (Bifet and Gavaldà, 2007) do not guarantee exchangeability or a deterministic recovery time. Under an explicitly justified total-variation bound on the *joint calibration–test law*, the probability of the same event changes by at most that bound; a marginal score-distance statement alone does not establish it. We neither estimate that distance nor claim recovery. Online labels selected by acceptance introduce additional selection bias. The runtime holds its threshold at zero on detected change until an explicit refresh; refresh remains an exploratory operation, not proof of safety.

**Protocol requirements beyond this study.** Workload, hardware and session groups are fixed before graph induction and assigned to disjoint train, retrieval, calibration and test roles; hashes of graphs, sources, prompts and calibrators are frozen before test results are opened, and a validator rejects overlap or unknown lineage. For each bundle we count proposed, rejected, vetoed, accepted, staged, completed and rolled-back outcomes, report gate TP/FP/FN/TN with both #unsafe-accepted/#unsafe and /#accepted, and leave unknown labels unknown; rollback precision needs independently labeled executed actions. A lower threshold accepts a subset of fixed candidates, but the unsafe fraction among survivors need not be monotone. Absolute per-workload metrics precede ratios against a named baseline; a Pareto claim needs several measured operating points per

Table 5: Follow-up selector replay: mean of ten test-run P95 values. Misses are observed lateness above 200 $\mu$s. Shared rows across selectors are intentional: selectors choosing the same configuration use the same held-out observations.

| Period | Selector | P95 ($\mu$s) | Misses |
|---|---|---|---|
| 5 | control | 179.7 | 13/400 |
| 5 | empirical min | 179.7 | 13/400 |
| 5 | graph min | 179.7 | 13/400 |
| 5 | seeded random | 811.6 | 248/400 |
| 8 | control | 155.7 | 13/400 |
| 8 | empirical min | 155.7 | 13/400 |
| 8 | graph min | 155.7 | 13/400 |
| 8 | seeded random | 642.2 | 225/400 |

method. Drift studies must keep the threshold trajectory and held-out outcomes at each decision, and faithfulness is tested by deleting cited and matched uncited context under fixed seeds.

# B Additional Measurements

## B.1 Selector Replay on the Follow-up

All controlled runs used one Intel Core i5-3570 host (four logical CPUs, Linux 6.17.0-23-generic). Each fresh child set its timer slack with `PR_SET_TIMERSLACK` and its affinity with `sched_setaffinity`, checked both by read-back, and recorded 40 wake-ups against absolute monotonic-clock targets after five warm-ups.

The empirical selector minimizes mean training-run P95 over the four configurations. A saturated two-factor representation encodes the same four means as a baseline, two main effects and their interaction; minimizing it adds no information to table lookup. Both selectors have the same 40 training runs and zero online search trials. An unchanged-control selector and a seeded uniform-random selector provide references. Choices are frozen before calibration and test. We evaluate these fixed selectors by paired *offline replay* on the same measured test grid, not by separate closed-loop executions. A block is the uncertainty unit, and paired reductions are calculated within each period against its explicit control.

Both informed selectors chose 50 $\mu$s slack and all CPUs, identical to control; their paired improvement is exactly zero in this replay. Control P95 is $179.7 \pm 102.0$ and $155.7 \pm 64.4$ $\mu$s at 5 and 8 ms (95% Student-$t$ half-width across ten blocks). The training interaction contrast is $-18.6$ $\mu$s; the pre-specified normalized-contrast bootstrap emitted no edge because its interval included zero. We do not turn a nonsignificant contrast into a dependency rule or a semantic-reasoning gain. Acquisition took 6.54 s for training, 8.29 s for retrieval and 10.15 s for calibration; graph construction and selection took 0.10 s, included in calibration-phase wall time.

Table 6: Matched-budget replay: % of 1,000 seeds choosing control / expected paired P95 change at 5 ms (%); last column: mean training evaluations of 1 ms settings at $B = 40$.

| Method | $B = 4$ | $B = 8$ | $B = 16$ | $B = 40$ | 1 ms evals |
|---|---|---|---|---|---|
| Random search | 62 / −55 | 74 / −38 | 85 / −22 | 98 / −3 | 19.8 |
| $\epsilon$-greedy (RL) | 65 / −51 | 86 / −20 | 95 / −8 | 99 / −1 | 3.8 |
| UCB1 (RL) | 65 / −51 | 77 / −33 | 91 / −12 | 99 / −1 | 12.0 |
| GP-EI (BO) | 65 / −51 | 83 / −25 | 95 / −8 | 99 / −1 | 2.5 |

## B.2 Matched-Budget Optimizers

Random search, $\epsilon$-greedy Q-learning ($\epsilon = 0.1$), UCB1 and GP-EI BO each evaluate $B \in \{4, 8, 16, 40\}$ recorded training runs, starting with one per configuration, and recommend their lowest observed or posterior mean, scored by the same replay over 1,000 seeds (Table 6). Control is best at both test periods, so this comparison can reveal only losses. With small budgets, noisy single-CPU runs mislead the methods; BO spends fewest evaluations on the 1 ms settings, whose training runs miss over 90% of deadlines.

## B.3 Gate Replay

## B.4 Staged Actuation

Across 80 fresh executions, 66 stop on a stage breach and 14 complete all stages; all 80 restore and read back both original values. At 5 and 8 ms, observed deadline misses are respectively $186/816$ and $190/880$; summed exposure durations are 4.106 and 7.066 s. Early stops reduce the number of observed events, so these fractions cannot be compared directly with fixed-duration, no-stop runs to claim an anomaly reduction. The stop criterion itself is not independent ground truth for rollback precision.

## B.5 Local Baseline Measurements

We executed three bounded microbenchmarks on the available host: an Intel Core i5-3570 with four logical CPUs, approximately 31.2 GiB RAM, and Linux 6.17.0-23-generic. These runs characterize the measurement pipeline: no kernel knob was modified and no LLM or controller was compared.

Each workload had 10 separate invocations of 100 measured operations, preceded by 10 warm-

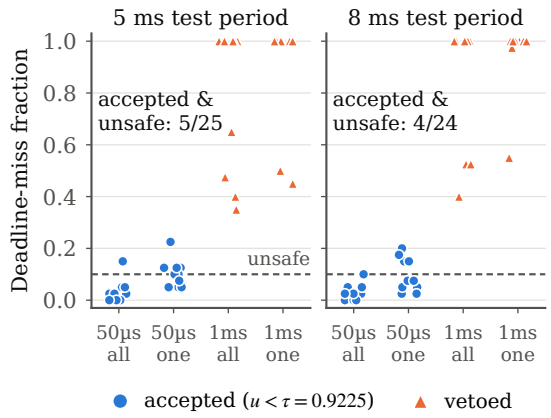



Figure 3: Frozen-gate replay on the 80 held-out test runs; each point is one run, grouped by timer slack (50 $\mu$s or 1 ms) and CPU affinity (all CPUs or one). Positive means unsafe (more than 10% of deadlines missed, dashed line) and a gate-positive decision is veto. Counts at every tested $\alpha$ (all give $\tau = .9225$): 5 ms, 20 accepted / 20 vetoed, TP 20, FP 0, FN 5, TN 15; 8 ms, 20 / 20, TP 20, FP 0, FN 4, TN 16. Blue points above the line are the accepted unsafe runs. Generated from raw logs by `plot_gate_replay.py`.

Table 7: Measured local baseline. Mean of run medians, and mean of run P95 values with 95% Student-$t$ CI half-width across 10 runs. Last column is the percentage of errors or operations exceeding a predeclared 50 ms deadline. No tuning gain is implied.

| Workload | Median ms | P95 ms | Events % |
|---|---|---|---|
| file_fsync | 1.903 | 2.035 ± 0.043 | 0.00 |
| sqlite_commit | 1.870 | 2.187 ± 0.273 | 0.00 |
| zlib_compress | 1.474 | 2.038 ± 0.356 | 0.00 |

up operations. Workload–run order was shuffled with seed 4088. SQLite inserts a 4 KiB value and commits each transaction in WAL mode with synchronous=FULL. The file test overwrites a 64 KiB region and calls fsync. The compression test compresses a fixed seeded 64 KiB payload at zlib level 6 and verifies decompression. Latency is wall-clock operation duration from a monotonic nanosecond clock; the compression measurement includes verification. The raw log retains every operation, run ID, deadline and error flag, with a SHA-256 hash and software/environment manifest.

The anomaly denominator is all measured operations, including failures; an operation counts once if either it fails or exceeds the deadline. This label is independent of veto and rollback. Zero events in these short runs is not evidence of zero population risk. We do not pool heterogeneous workload latencies or infer kernel tuning effects from these baseline timings.

Table 8: Held-out kernel intervention measurements. P95 is the mean of ten run-level P95 wake-up latenesses. Misses count lateness exceeding the predeclared 200 $\mu$s deadline. CPU affinity is all four CPUs or one CPU.

| Period | Slack/CPUs | P95 ($\mu$s) | Misses |
|---|---|---|---|
| 2 | 50us/all | 89.5 | 2/800 |
| 2 | 50us/one | 109.4 | 18/800 |
| 2 | 1ms/all | 1016.8 | 800/800 |
| 2 | 1ms/one | 968.9 | 771/800 |
| 5 | 50us/all | 94.2 | 2/800 |
| 5 | 50us/one | 133.8 | 22/800 |
| 5 | 1ms/all | 1022.6 | 663/800 |
| 5 | 1ms/one | 1073.7 | 680/800 |

### B.6 Temporal-Holdout Kernel Interventions

We additionally executed real Linux timer-slack and CPU-affinity interventions in fresh owned child processes, without modifying host-wide sysctls. The two factors were timer slack (50 $\mu$s or 1 ms, set by PR_SET_TIMERSLACK) and affinity (all four available CPUs or CPU 0). Every child checked kernel readback before recording events. This isolated measurement driver is separate from the dry-run GroundKern actuator.

For each periodic wake-up interval (2 or 5 ms), five training blocks and ten subsequent test blocks evaluated all four configurations. Within each block, the eight interval/configuration cases were randomly ordered using seed 4088. Each fresh process performed ten warm-up and 80 measured wake-ups against absolute monotonic-clock targets. The plan was written before execution; the configuration minimizing mean training-run P95 lateness was frozen before test execution. This is temporal holdout within one host and workload family, not the independent workload/hardware split required for claims of generalization. The complete 120-process experiment retained 9,600 event timestamps, readbacks, frozen selections and hashes. No external competing load was introduced.

The training selector chose the explicit control (50 $\mu$s slack, all CPUs) at both intervals: there is no selected-policy improvement over that control. Its test P95 was $89.5 \pm 9.2$ and $94.2 \pm 6.9$ $\mu$s at 2 and 5 ms, respectively (95% Student-$t$ half-width across ten runs). Raising slack to 1 ms with all CPUs increased P95 by $927.3 \pm 16.4$ and $928.4 \pm 10.0$ $\mu$s in paired blocks. Affinity-only differences were $19.9 \pm 36.7$ and $39.6 \pm 62.6$ $\mu$s; these intervals include zero. The slack effect is consistent with

timer coalescing; no power measurement, graph discovery, LLM advantage or staged rollback is inferred. A miss is defined by observed wake-up time, independently of any safety gate.

## C Prompt and Output Schema

Every model receives the following system prompt:

> Select one allowed config to minimize periodic wake-up P95 latency. Return JSON with config (exact allowed name), cited_ids (evidence IDs), and explanation. Evidence contains measured training summaries and retrieval runs, not held-out test outcomes. Do not invent evidence.

The user message is a JSON object with the four allowed configuration names and the evidence list for the input variant. Items have the forms

```
{"id": "training-table", "kind":
"table", "values": {"s50000_aall":
132.9413, ...}}
{"id": "interaction", "kind": "graph",
"contrast_us": -18.62, "edges": [],
"scope": "same measured table; graph is
a reparameterization, not extra
evidence"}
{"id": "retrieval-3-0-s50000_aall",
"kind": "retrieval", "config":
"s50000_aall", "period_ns": 3000000,
"p95_us": 126.36}
```

The output schema requires `config` from the allowed list, `cited_ids` with at most six entries restricted to the supplied IDs (none for the no-evidence variant), and an `explanation` of at most 400 characters. Local models enforce it as a llama.cpp JSON grammar with greedy decoding; hosted models receive it as a strict JSON schema through OpenRouter, restricted to providers that honor it. Every answer is re-validated, and an invented or deleted citation makes it invalid; the explanation length is not re-checked, and 14 of 20 DeepSeek-V3.2 answers exceed 400 characters. Claim and citation counts in §5 are recomputed from these records by `claim_taxonomy.py` and `citation_analysis.py`. A single script, `reproduce_paper.py`, reruns every analysis on a copy of the raw logs and checks that each table and quoted number in the paper is reproduced exactly; the supplementary `ARTIFACTS.md` maps each result to its data and script and includes a data statement. The supplementary records contain each prompt, its SHA-256 hash, the raw response and, for hosted models, the routed provider.

**Claim-label checks and intervals.** The 97 claim labels were drafted with an AI assistant. A rule-based check (`claim_label_autocheck.py`) re-derives the expected label of 56 claims directly from the evidence (values attributed to the training table, lists of runs, superlatives, ranges, thresholds, units and no-evidence claims) and agrees with every drafted label; flipping each label between *correct* and *wrong* is detected in 76 of 97 cases. The remaining 41 claims require judgment. One author then reviewed every label against the evidence, reading these 41 claims in full, and changed none (agreement with the draft 100%); because the reviewer checked an existing draft, this is a confirmation rather than an independent annotation. For agreement, two further annotators who are not authors each labeled all 97 claims blind and independently, from the claims, their source sentences and written guidelines, without the drafted labels, checks or each other's labels. The first agrees with the verified labels on 94.8% of claims over the seven labels (Cohen's $\kappa = 0.918$) and 95.9% for misstated versus not ($\kappa = 0.914$); all five disagreements are claims this annotator judged correct (three "significantly lower" statements, one partly wrong claim, one range of cited runs). The second agrees on 90.7% ($\kappa = 0.856$) and 90.7% ($\kappa = 0.808$); five of its nine disagreements read "lowest across all retrieval runs (runs 0–5)" as restricted to runs 0–5, and two arise because the blind sheet showed a sentence from another seed for the claim (since fixed in `claim_review_agreement.py`). Over the verified labels and both annotators, Fleiss' $\kappa$ is 0.847 (seven labels) and 0.814 (misstated versus not), and the majority label equals the verified label for all 97 claims. The reported counts use the verified labels; with the annotators' labels, 37 and 38 rather than 41 of 97 claims are misstated. With 95% Wilson intervals, the misstated proportions are 57% (39–73%) for local models, 36% (26–48%) for hosted models and 42% (33–52%) overall; 97% (94–98%) of cited retrieval runs belong to the chosen configuration and 80% (75–85%) come from the first half of the evidence list. These intervals treat claims and citations as independent, which overstates their precision because hosted explanations repeat claims across seeds.

**Entailment-based attribution judges.** To test whether citation-support judges catch these errors, `entailment_judge.py` gives each of the 93 evidence-based claims to eight judges with pinned revisions: three NLI cross-encoders (DeBERTa-

v3-large trained on five NLI sets, DeBERTa-v3-large trained on SNLI/MNLI, RoBERTa-large-MNLI; *supported* = entailment), Flan-T5-large asked whether the claim is fully supported (*supported* = yes), and four judges trained to fact-check claims against grounding documents, AlignScore-large (Zha et al., 2023) and MiniCheck with RoBERTa-large, DeBERTa-v3-large and Flan-T5-large backbones (Tang et al., 2024) (*supported* = score above 0.5). As in citation-support evaluation, the premise is the verbalized evidence the model cited; the hypothesis is either the explanation sentence that states the claim or the extracted claim. Table 9 compares the judges with the author-verified labels. No judge separates misstated from correct claims well (balanced accuracy 0.47–0.68), and at least one judge accepts 28 of the 41 misstated claims. Lenient judges accept up to half of the wrong values, ranges and units, the near-misses that quote correct numbers; strict judges reject nearly all overstatements but also most correct claims. The fact-checkers are the strictest: MiniCheck accepts at most 3 of 51 correct claims. Correct claims about uncited evidence cannot be entailed by the cited items alone, but giving the two long-context MiniCheck judges all 42 evidence items as premise (claim*) still leaves them near chance (0.49 and 0.52), because claims such as ranges and orderings require aggregating many items. TRUE (T5-11B) did not fit the 15 GB of memory of our CPU container and was not tested.

**Compute budget.** Local models ran as 4-bit GGUF files (Q4_K_M; Phi-4 Q4_K) with llama-cpp-python 0.3.35 (`https://github.com/abetlen/llama-cpp-python`) on a four-core CPU container without a GPU, with an 8,192-token context and at most 384 output tokens. Their 100 calls took 11,580 s of wall time (about 3.2 CPU-hours): Qwen2.5-7B 1,603 s, Llama-3.1-8B 2,325 s, Qwen2.5-14B 3,136 s and Phi-4 4,516 s. The 150 hosted calls through OpenRouter, with at most 4,096 output tokens, took 1,224 s in total and cost about 0.5 USD; the 150 robustness calls (plus two retries) cost 0.83 USD over about 2.5 h of wall time. The eight attribution judges ran on the same CPU container. Kernel measurements ran on the Intel Core i5-3570 host described in Appendix B; acquiring the training, retrieval and calibration roles took 6.54, 8.29 and 10.15 s. Optimizer replays over 1,000 seeds take seconds on one CPU.

Table 9: Off-the-shelf attribution judges on the 93 evidence-based claims: claims accepted as supported among wrong values (wrong, wrong unit, wrong attribution; 22), overstatements (19) and correct claims (51), with balanced accuracy (BA) against the verified labels. Hyp.: hypothesis is the source sentence or the extracted claim.

| Judge | Hyp. | Wrong | Over. | Correct | BA |
|---|---|---|---|---|---|
| DeBERTa-v3 (5 NLI sets) | sent. | 7/22 | 3/19 | 16/51 | 0.54 |
| DeBERTa-v3 (5 NLI sets) | claim | 11/22 | 1/19 | 29/51 | 0.64 |
| DeBERTa-v3 (SNLI/MNLI) | sent. | 3/22 | 0/19 | 11/51 | 0.57 |
| DeBERTa-v3 (SNLI/MNLI) | claim | 3/22 | 0/19 | 19/51 | 0.65 |
| RoBERTa-large-MNLI | sent. | 1/22 | 0/19 | 4/51 | 0.53 |
| RoBERTa-large-MNLI | claim | 3/22 | 0/19 | 13/51 | 0.59 |
| Flan-T5-large | sent. | 6/22 | 8/19 | 18/51 | 0.51 |
| Flan-T5-large | claim | 9/22 | 4/19 | 34/51 | 0.68 |
| AlignScore-large | sent. | 5/22 | 0/19 | 12/51 | 0.56 |
| AlignScore-large | claim | 6/22 | 0/19 | 12/51 | 0.54 |
| MiniCheck-RoBERTa | sent. | 2/22 | 0/19 | 1/51 | 0.48 |
| MiniCheck-RoBERTa | claim | 3/22 | 0/19 | 3/51 | 0.49 |
| MiniCheck-DeBERTa | sent. | 1/22 | 0/19 | 0/51 | 0.49 |
| MiniCheck-DeBERTa | claim | 3/22 | 0/19 | 1/51 | 0.47 |
| MiniCheck-Flan-T5 | sent. | 1/22 | 0/19 | 0/51 | 0.49 |
| MiniCheck-Flan-T5 | claim | 2/22 | 0/19 | 0/51 | 0.48 |
| MiniCheck-DeBERTa | claim* | 5/22 | 0/19 | 5/51 | 0.49 |
| MiniCheck-Flan-T5 | claim* | 5/22 | 0/19 | 8/51 | 0.52 |

## D Results by Model Scale and Family

`scale_family_analysis.py` re-analyses the existing records by scale and family without new model calls (Table 10). It uses only full-context answers, since local models were also audited on ablated prompts; claims of local models are counted once per distinct explanation because their greedy outputs repeat across seeds, whereas hosted explanations vary and repeat claims across seeds. Among the eight open-weight models, the misstated share falls with log total parameters (Spearman $\rho = -0.78$, $p = 0.02$), and local models misstate more often than hosted open-weight models (7/8 versus 21/52; Fisher's exact test $p = 0.02$). With eight local claims and repeated hosted claims, this is a trend, not an estimate. The error types change with scale: wrong units and source attributions occur only in 7–15B models, whereas overstatements and wrong values occur at every scale and in every family. Citation behavior does not track scale: every open-weight model cites only runs of its chosen configuration; Gemini-Pro is the only model that cites runs of other configurations (8 of 21), and front-loading is strongest in local models (100 of 110 cited runs in the first half) and weakest in Llama-70B (14 of 28). Within the Qwen and Llama families, the larger model misstates a smaller share (Qwen 2/2 and 2/2 at 7B and 14B versus 3/6 at 235B; Llama 2/2 versus 2/7).

Table 10: Full-context results by model: answers choosing the selector's configuration (Sel.), cited retrieval runs of the chosen configuration and from the first half of the evidence list, misstated claims, and misstated claims by type (overstated / wrong value / wrong unit or source). Gemini parameter counts are undisclosed.

| Model | Params | Sel. | Cited chosen | First half | Misst. | Over./Val./Unit |
|---|---|---|---|---|---|---|
| Qwen-7B | 7.6B | 5/5 | 25/25 | 25/25 | 2/2 | 1/0/1 |
| Llama-8B | 8B | 0/5 | 30/30 | 25/30 | 2/2 | 0/2/0 |
| Qwen-14B | 14.7B | 5/5 | 25/25 | 25/25 | 2/2 | 1/0/1 |
| Phi-4 | 14.7B | 5/5 | 30/30 | 25/30 | 1/2 | 1/0/0 |
| *Local open, 7–15B* | | *15/20* | *110/110* | *100/110* | *7/8* | *3/2/2* |
| Llama-70B | 70.6B | 5/5 | 28/28 | 14/28 | 2/7 | 0/2/0 |
| Qwen3-235B | 235B | 5/5 | 30/30 | 25/30 | 3/6 | 3/0/0 |
| DeepSeek-V3.2 | 671B | 5/5 | 25/25 | 25/25 | 6/16 | 2/4/0 |
| Nemotron-550B | 550B | 5/5 | 30/30 | 25/30 | 10/23 | 5/5/0 |
| *Hosted open, 70–671B* | | *20/20* | *113/113* | *89/113* | *21/52* | *10/11/0* |
| Gemini-Flash | – | 5/5 | 16/16 | 10/16 | 2/12 | 2/0/0 |
| Gemini-Pro | – | 5/5 | 13/21 | 10/21 | 2/5 | 2/0/0 |
| *Proprietary* | | *10/10* | *29/37* | *20/37* | *4/17* | *4/0/0* |

Table 11: Hosted models under prompt, order, cap and decoding variations (30 answers per condition): answers choosing the selector's configuration (Sel.), mean citations per answer, cited retrieval runs of the chosen configuration, in the first half of the list as presented, and from blocks 0–4, and explanations with a quantifier overstatement (automatic proxy).

| Condition | Sel. | Cites | Chosen | First half | Blocks 0–4 | Overst. |
|---|---|---|---|---|---|---|
| Original (greedy) | 30/30 | 5.5 | 142/150 | 109/150 | 109/150 | 21/30 |
| Paraphrased prompt | 30/30 | 4.8 | 131/131 | 94/131 | 94/131 | 15/30 |
| Reversed runs | 30/30 | 5.1 | 138/138 | 88/138 | 50/138 | 13/30 |
| Shuffled runs | 30/30 | 5.1 | 138/138 | 76/138 | 67/138 | 9/30 |
| No citation cap | 30/30 | 9.0 | 256/256 | 128/256 | 128/256 | 21/30 |
| Sampling ($T$=0.7) | 30/30 | 5.1 | 138/138 | 94/138 | 94/138 | 13/30 |

## E Robustness to Prompt, Evidence Order, Citation Cap and Decoding

`prompt_robustness.py` reruns the full-context prompt of the six hosted models, five seeds each, with the same frozen evidence, strict JSON schema and OpenRouter settings as the main audit, under five conditions: a paraphrased system prompt; retrieval runs in reverse order; retrieval runs in a seed-specific random order; no citation cap (any number of supplied IDs); and sampling at temperature 0.7. All 150 answers are valid after one retry of two answers that a provider truncated. Table 11 reports the results. Every answer in every condition chooses the selector's configuration, and every cited run belongs to the chosen configuration (in the original runs only Gemini-Pro cites others). Reversing the runs moves citations with list position for four models (Llama-70B, Qwen3-235B and both Gemini models), whereas Nemotron keeps citing blocks 0–4 and DeepSeek-V3.2 falls in between; without the cap, four models cite all ten runs of the chosen configuration. The explanations of these new answers have no verified claim labels, so we score them with two automatic proxies, applied identically to the original runs: a universal quantifier attached to a comparison ("consistently the lowest", "lower in all runs") or "significantly lower", which the evidence does not support; and decimal numbers that match no evidence value or derived minimum, maximum, mean or difference. The quantifier overstatement recurs in every condition (9 to 21 of 30 explanations, 21 in the original runs), while unsupported numbers are rare (at most two per condition), consistent with the verified finding that misstated claims mostly quote correct numbers.